# Bhavesh Singh Tomer

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## EDUCATION

**IIT (ISM) Dhanbad** *Expected May 2027*
*B.Tech in Mining Machinery Engineering (Department of Mechanical Engineering), CGPA: 8.00* *Dhanbad, Jharkhand*

- **Relevant Coursework:** Data Structures and Algorithms (C++), Digital Image Processing, Computer Programming, Computer Programming Lab

## EXPERIENCE

**1Stop** *Apr 2024 – Jun 2024*
*Web Development Front-End*

- Developed responsive websites using HTML, CSS, JavaScript, Bootstrap, and jQuery, with mobile-friendly layouts and reusable UI components.
- Implemented interactive features such as navigation menus, form validation, dynamic content updates, sliders, carousels, and smooth scrolling to improve user experience.

## PROJECTS

**HistoLens AI: High-Resolution Digital Pathology & WSI Analysis | *Python, FastAPI, OpenSeadragon, OpenCV, Scikit-Image, DeepZoom*** *[GitHub]*

- Architected an on-demand DeepZoom tile streaming pipeline (256×256 tiles) to process gigapixel Whole Slide Images (~50K×50K), decoupling multi-gigabyte files from RAM to sustain <150 MB runtime memory.
- Engineered an automated tissue segmentation module using Otsu and HSV chroma thresholding to filter 80% blank glass background, coupled with Beer-Lambert optical density deconvolution for Hematoxylin & Eosin stain separation.
- Implemented a high-magnification (40×) ROI patch analyzer with single-cell nuclei segmentation, quantifying nuclear crowding (>2500 cells/mm²), pleomorphism, and malignancy probability in real time.
- Developed a full-stack digital pathology viewer with OpenSeadragon (1×–40× multi-scale zoom), real-time tumor probability heatmap overlays, WHO disease stratification, TSR calculation, and automated signed PDF report generation.

**AgenticAI RAG for SWE-QA | *LangGraph, LangChain, Qdrant, Hugging Face, FastAPI, Docker, RAGAS*** *[GitHub]*

- Developed an Agentic Retrieval-Augmented Generation (RAG) system for software engineering question answering over technical documentation and open-source code repositories using LangGraph, Qdrant, and Mistral 7B.
- Implemented Adaptive-RAG with intelligent query routing and Corrective-RAG featuring retrieval evaluation, query refinement, and iterative re-retrieval, improving Answer Relevancy by 17.8% over a baseline RAG pipeline.
- Assembled a hybrid retrieval pipeline combining BM25 and dense vector search (Sentence Transformers + Qdrant) with BGE reranking and AST-aware code chunking across multiple repositories.
- Containerized the application using Docker, exposed FastAPI REST APIs, and evaluated using RAGAS, achieving 91.8% Faithfulness, 89.6% Context Precision, 93.1% Answer Relevancy, and an average response latency of 840 ms.

**Bitcoin x402 Payment System | *Starknet, Next.js, TypeScript, Express.js, Web3*** *[GitHub]*

- Developed an HTTP 402-based payment system enabling Bitcoin-denominated micropayments for API access and pay-per-use services on Starknet.
- Built backend workflows for HTTP 402 payment handling, transaction verification, payment validation, and secure request processing.
- Developed a Next.js payment interface with QR-based wallet integration, real-time transaction tracking, and payment diagnostics.
- Integrated blockchain payment workflows with nonce-based transaction validation and explored privacy-preserving ZK proof mechanisms for payment verification.

## TECHNICAL SKILLS

**Languages:** Python, C++, JavaScript, TypeScript, Solidity, SQL
**Technologies/Frameworks:** FastAPI, OpenCV, PyTorch, Scikit-Image, OpenSeadragon, React.js, Express.js, Node.js, Docker, MongoDB
**Concepts:** Digital Image Processing, Computer Vision, DeepZoom/WSI Tiling, Agentic RAG, Blockchain, REST APIs

## ACHIEVEMENTS

- Team member of a winning project in ETHGlobal's Agentic Ethereum – OPSEC track.

## SOCIAL ENGAGEMENTS

**Club Member:** Cyber Labs – Tech Society of IIT Dhanbad
**Volunteer:** Srijan (Cultural Fest) Documentation Team
**Lawn Tennis:** College Team
**Captain:** House Captain at DPS Bareilly